

MINI-ASSESSMENT: MULTI-STEP PROMPT WORKFLOW

Transforming Raw Placement-Preparation Notes into a Structured Interview Preparation Kit

STUDENT DETAILS	INFORMATION
Name	CH. Siva Mowny
Register Number	99230040092
Year / Branch	IV Year / CSE-AIML
Assessment Type	Chained Multi-Step AI Prompt Workflow

Raw Notes → Planning → Interview Kit → Critique & Refinement → Final Output

1. Scenario Selected

Technical / Placement Preparation

The scenario demonstrates how unstructured placement-preparation notes can be transformed into a clear, practical and interview-ready preparation kit through a three-step chained AI workflow.

2. Objective

To use a multi-step prompting process to organize raw placement notes, generate a structured interview preparation kit, and then evaluate and refine the generated content against explicit quality constraints.

3. Raw Placement-Preparation Notes

The following notes are the starting material for the workflow:

- Placement preparation is focused on aptitude, communication, programming and technical interviews.
- Important aptitude areas include Number System, Percentages, Profit & Loss, Ratio & Proportion, Time & Work, Time-Speed-Distance, Averages, Probability, Permutations & Combinations and Data Interpretation.
- Technical preparation includes Java programming, OOP concepts, DSA, SQL, DBMS, AI/ML fundamentals and project explanation.
- Java topics include variables, data types, operators, conditions, loops, arrays, strings, methods, classes, objects, inheritance, polymorphism, encapsulation and abstraction.
- DSA preparation includes arrays, strings, linked lists, stacks, queues, trees, searching, sorting and problem-solving.

- SQL preparation includes SELECT, WHERE, GROUP BY, ORDER BY, JOINS, aggregate functions, subqueries and basic database concepts.
- AI/ML preparation includes supervised and unsupervised learning, classification, regression, CNNs and common ML evaluation concepts.
- Interview preparation should include Tell Me About Yourself, project explanation, strengths, weaknesses, career goals and company-related questions.
- Preparation should be organized by priority so that high-value topics can be revised first.

4. Workflow Overview

Step	Purpose	Dependency
Step 1	Extract topics, priorities and interview requirements from raw notes.	Uses only raw notes.
Step 2	Generate a structured interview preparation kit.	Must use Step 1 output.
Step 3	Critique, correct and refine the kit.	Must evaluate Step 2 output against explicit constraints.

5. Step 1 — Extraction & Planning

Goal: Convert the raw notes into a preparation blueprint that can directly guide the next AI response.

Prompt 1 — Exact Prompt

You are a placement-preparation planning assistant. Analyze the raw placement-preparation notes provided above and convert them into a structured preparation blueprint for a final-year CSE-AIML student.

Extract:

1. Major preparation categories.
2. Topics under each category.
3. High-priority topics that should be revised first.
4. Interview questions or areas that need separate preparation.
5. A logical order for preparing the material.

Do not create the final preparation kit yet. Do not invent certifications, company experiences, interview results or achievements. Keep the plan practical and easy to convert into a study kit.

AI Response — Step 1 Output

Category	Extracted Topics	Priority
Aptitude	Number System; Percentages; Profit & Loss; Ratio; Time & Work; Time-Speed-Distance; Averages; Probability; P&C; DI	High
Programming / Java	Syntax; variables; operators; conditions; loops; arrays;	High

	strings; methods; classes; objects	
OOP	Encapsulation; inheritance; polymorphism; abstraction	High
DSA	Arrays; strings; linked lists; stacks; queues; trees; searching; sorting; problem solving	High
SQL / DBMS	SELECT; WHERE; GROUP BY; ORDER BY; JOINS; aggregate functions; subqueries; database basics	High
AI/ML	Supervised vs unsupervised learning; classification; regression; CNN; evaluation concepts	Medium-High
HR / Communication	Tell Me About Yourself; strengths; weakness; goals; company questions; communication	High
Project Discussion	Project objective; technology used; role; challenges; results; future improvements	High

Planning conclusion: The final kit should begin with high-frequency interview foundations, followed by coding/DSA/SQL, AI/ML, aptitude revision and finally HR/project practice. The kit should also contain quick revision material and practice questions.

6. Step 2 — Generation / Drafting

Goal: Transform the Step 1 blueprint into a usable interview preparation kit. This step explicitly depends on the previous output.

Prompt 2 — Exact Prompt

Using the preparation blueprint you created in Step 1, generate a structured Placement Interview Preparation Kit.

The kit must include:

1. A prioritized preparation roadmap.
2. Java and OOP revision points.
3. DSA topics with practice focus.
4. SQL/DBMS revision checklist.
5. AI/ML revision checklist.
6. Aptitude formula/revision areas.
7. HR interview questions with preparation guidance.
8. Project-explanation checklist.
9. A final quick-revision checklist.

Use the exact topics extracted in Step 1. Do not add unsupported personal achievements or claim that any interview was cleared. Keep the content concise enough for revision but useful enough for interview preparation.

AI Response — Step 2 Output

A. Prioritized Roadmap

Priority	Preparation Area	Action
1	Java + OOP	Revise concepts and solve basic coding questions.
2	DSA	Practice arrays, strings, linked lists, stacks, queues, trees, searching and sorting.
3	SQL / DBMS	Practice queries, joins, grouping, aggregates and subqueries.
4	Project + HR	Prepare self-introduction, project explanation, strengths, weakness and goals.
5	AI/ML	Revise supervised/unsupervised learning, classification, regression and CNN basics.
6	Aptitude	Revise formulas and solve timed practice sets.

B. Java & OOP Quick Revision

- Java basics: variables, data types, operators, conditions, loops, arrays, strings and methods.
- Class: a blueprint used to create objects. Object: an instance of a class.
- Encapsulation: bundling data and methods and controlling access.
- Inheritance: deriving a new class from an existing class.
- Polymorphism: allowing one interface or method concept to have multiple forms.
- Abstraction: exposing essential behavior while hiding implementation details.

C. DSA Practice Focus

- Arrays: traversal, searching, insertion/deletion logic and common problem patterns.
- Strings: frequency counting, reversal, palindrome and character processing.
- Linked Lists: insertion, deletion, traversal and basic pointer manipulation.
- Stacks & Queues: implementation and common use cases.
- Trees: traversal concepts including inorder, preorder and postorder.
- Searching & Sorting: understand basic algorithms and their use cases.

D. SQL / DBMS Checklist

- Practice SELECT, WHERE, ORDER BY and GROUP BY.
- Understand aggregate functions such as COUNT, SUM, AVG, MIN and MAX.
- Practice INNER JOIN and other common JOIN concepts.
- Revise subqueries and basic database terminology.

E. AI/ML Checklist

- Differentiate supervised and unsupervised learning.
- Revise classification and regression.
- Understand the basic idea of CNNs for image-related tasks.
- Revise common model evaluation concepts and why evaluation matters.

F. HR + Project Questions

Question / Area	Preparation Focus
Tell me about yourself	Name, degree, CSE-AIML background, technical skills, projects and career interest.
Explain your project	Problem, objective, technologies, personal contribution, challenges and outcome.
What are your strengths?	Choose genuine strengths and support them with examples.
What is one weakness?	Give an honest, manageable weakness and explain improvement.
Why should we hire you?	Connect technical preparation, problem solving, learning ability and role requirements.
Where do you see yourself?	Show realistic growth and willingness to learn.

G. Final Quick-Revision Checklist

- ☐ Java basics and OOP revised
- ☐ DSA core topics revised
- ☐ SQL query practice completed
- ☐ AI/ML concepts revised
- ☐ Aptitude formulas revised
- ☐ Self-introduction practiced
- ☐ Project explanation practiced
- ☐ HR questions practiced
- ☐ Company and role researched

7. Step 3 — Evaluation & Refinement Gate

Goal: Review the Step 2 output against specific constraints before accepting it as the final deliverable.

Prompt 3 — Exact Prompt

Critically evaluate the interview preparation kit you generated in Step 2.

Check it against these requirements:

1. Does it cover the important categories extracted in Step 1?
2. Are high-priority topics clearly prioritized?
3. Does every major section trace back to the Step 1 blueprint?
4. Are Java/OOP, DSA, SQL/DBMS, AI/ML, aptitude, HR and project preparation covered?
5. Is the content useful for quick interview revision?

6. Are any important raw-note topics missing?
7. Did the response introduce unsupported personal achievements or claims?
8. Are the HR and project sections practical?
9. Is the organization clear and easy to follow?
10. Is the final checklist actionable?

First list the weaknesses you find. Then correct them. Finally provide the refined final interview preparation kit. Do not invent personal facts.

AI Response — Critique

Check	Evaluation	Refinement
Coverage	Most major categories are covered.	Retain all Step 1 categories explicitly.
Priority	Priority order is present.	Keep Java/OOP, DSA and SQL highly visible.
Traceability	Sections correspond to the planning blueprint.	Keep the same category names where possible.
Technical coverage	Java, OOP, DSA, SQL and AI/ML are included.	Add a compact revision checklist for each.
HR / Project	Questions are practical but can be more structured.	Add answer structure and project explanation sequence.
Aptitude	Topics are listed but formulas are not detailed.	Keep this as a revision-area checklist rather than inventing formulas not present in the raw notes.
Unsupported claims	No achievements or interview results are claimed.	Maintain this constraint.
Actionability	Checklist is useful.	Add a final mock-interview and timed-practice reminder.

Refined Final Interview Preparation Kit

1. Preparation Order

- First: Java, OOP and basic coding.
- Second: DSA problem solving and common data structures.
- Third: SQL/DBMS query practice.
- Fourth: Project explanation and HR communication.
- Fifth: AI/ML revision.
- Sixth: Aptitude formula revision and timed practice.

2. Interview Answer Framework

Area	Answer Structure
Self Introduction	Background → CSE-AIML → technical skills → projects → role interest
Project	Problem → objective → technologies →

	contribution → challenge → outcome → improvement
Technical Question	Define → explain simply → give example → mention use case
HR Question	Direct answer → short reason/example → positive conclusion

3. Final Readiness Check

- ☐ Can explain Java and OOP concepts without memorizing definitions.
- ☐ Can solve basic DSA coding problems.
- ☐ Can write and explain common SQL queries.
- ☐ Can explain supervised vs unsupervised learning and classification vs regression.
- ☐ Can explain a project clearly from problem to outcome.
- ☐ Can deliver a confident self-introduction.
- ☐ Can answer common HR questions naturally.
- ☐ Can complete aptitude practice within a time limit.
- ☐ Can identify areas that need last-minute revision.
- ☐ Can participate in a mock interview.

8. Live Chat / Share-Link Proof

The assessment requires the prompt chain to be tested live in the assigned AI interface. Insert actual screenshots from the assigned interface in the spaces below.

Screenshot 1 — Step 1 Prompt + AI Response

1. Step 1 – Prompt + AI Response (Extraction & Planning)

ChatGPT

New chat

Search

Library

You

I have raw placement-preparation notes. Please analyze the notes and extract the key topics, group them into main categories, and create a structured plan for my interview preparation. Include important subtopics. My notes cover aptitude, Java, DSA, SQL, AI/ML, HR and project preparation. Also suggest a study timeline for 4 weeks.

ChatGPT

Here is the extracted and structured plan based on your placement preparation notes:

1. Key Topics Extracted and Categorized

Category	Key Topics / Subtopics	Priority Level
Aptitude	Number System, Time & Work, Percentage, Profit & Loss, Ratio & Proportion, Data Interpretation, Logical Reasoning	High
Java & OOP	Basics, OOP concepts, Collections, Exception Handling, Multithreading	High
DSA	Arrays, Strings, Linked List, Stack, Queue, Trees, Graphs, Sorting, Searching	High
SQL	Joins, Subqueries, Aggregate Functions, Window Functions, Indexing, Query Optimization	Medium
AI/ML	ML Basics, Supervised/Unsupervised Learning, Python, Libraries (NumPy, Pandas, Scikit-learn), Projects	Medium
HR	Common Questions, Self Introduction, Behavioral Questions, Company Culture	Medium
Project Preparation	Domain Knowledge, Technical Explanation, Resume Shortlisting, Past Projects	Medium

4-Week Study Timeline

- Week 1: Aptitude + Java Basics + SQL Fundamentals
- Week 2: DSA Practice + Advanced Java + SQL Queries
- Week 3: AI/ML Concepts + Project Preparation + HR Questions
- Week 4: Mock Interviews + Revision + Weak Area Improvement

Additional Suggestions

- Use LeetCode/GFG for DSA practice
- Build 2-3 real-world projects (AI/ML + Web)
- Practice coding on a daily basis (1-2 hours)
- Prepare 1-minute self-introduction
- Keep a notebook for formulas and key concepts

Evidence note: The screenshot should clearly show the prompt and corresponding AI output.

Screenshot 2 — Step 2 Prompt + AI Response



Evidence note: The screenshot should clearly show the prompt and corresponding AI output.

Screenshot 3 — Step 3 Critique + Refined Output



Evidence note: The screenshot should clearly show the prompt and corresponding AI output.

9. Chaining Dependency Proof

Step	Uses Previous Output?	Evidence
Step 1	No — starts from raw placement notes.	Creates the preparation blueprint.
Step 2	Yes — uses Step 1 blueprint.	Transforms extracted categories into the preparation kit.
Step 3	Yes — evaluates Step 2 output.	Critiques and refines the generated kit.

10. Critique Gate Proof

Step 3 is a mandatory evaluation gate rather than a repeat generation prompt. It checks coverage, traceability, unsupported claims, organization and actionability before producing the refined final version.

11. Final Submission Checklist

- ☐ Student name and register number are included.
- ☐ Scenario and objective are clearly stated.

- ☐ Raw placement notes are shown.
- ☐ At least three distinct chained prompts are included.
- ☐ Prompt 2 explicitly depends on Step 1.
- ☐ Prompt 3 explicitly evaluates Step 2.
- ☐ Intermediate AI outputs are documented.
- ☐ Critique/refinement output is documented.
- ☐ Actual live-interface screenshots are inserted.
- ☐ Final document is reviewed for clarity and completeness.

12. Conclusion

This workflow demonstrates a complete chained prompting process: raw placement-preparation notes are first converted into a structured blueprint, the blueprint is then transformed into an interview preparation kit, and the kit is finally evaluated and refined using explicit quality criteria. The process is therefore more than a single-shot generation task and provides clear evidence of dependency between the three AI steps.